Integral Inequalities, Symmetry, and the Beta Function

N-20260106

Integral inequalities become especially transparent when one interprets integration as averaging. Convexity compares the average of a function with its values at special points; symmetry replaces an integral with another integral of exactly the same value; and repeated integration by parts produces recurrences whose limiting behavior is encoded by special functions. We will use these three mechanisms to connect quadrature errors, a sharp reciprocal inequality, and Wallis' formula.[1]

§1.1 Hermite–Hadamard is a theorem about averages

Suppose f is convex on $[a,b]$, $a<b$, and put $m=(a+b)/2$. Convexity gives

$$f(m)\le \frac{f(x)+f(a+b-x)}{2}.$$

Integrate in x to obtain the lower bound in

$$\boxed{f(m)\le \frac{1}{b-a}\int_a^b f(x)\,dx\le \frac{f(a)+f(b)}{2}.}$$

For the upper bound, use the fact that a convex function lies below its endpoint chord:

$$f(x)\le \frac{b-x}{b-a}f(a)+\frac{x-a}{b-a}f(b).$$

Integrating the chord gives the endpoint average.

The two inequalities measure different approximations to the same integral. In numerical analysis they are the midpoint and trapezoidal rules, each exact for affine functions. The sign of the quadrature error is already determined by convexity.

§1.2 Quadrature errors are weighted second derivatives

Let $f\in C^2([a,b])$. Twice integrating by parts gives the exact identity

$$\frac{b-a}{2}(f(a)+f(b))-\int_a^b f(x)\,dx=\frac{1}{2}\int_a^b (x-a)(b-x)f''(x)\,dx.$$

The kernel $(x-a)(b-x)$ is nonnegative. By the integral mean value theorem there is $\xi\in(a,b)$ such that

$$\boxed{\frac{f(a)+f(b)}{2}-\frac{1}{b-a}\int_a^b f=\frac{(b-a)^2}{12}f''(\xi).}$$

[1] For the beta–gamma identity, see NIST Digital Library of Mathematical Functions, §5.12: https://dlmf.nist.gov/5.12.

For the midpoint approximation, expand around m with the integral remainder. The odd first-order term integrates to zero, and the remaining second-order remainder has a nonnegative kernel. Another application of the integral mean value theorem yields

$$\boxed{\frac{1}{b-a}\int_a^b f - f(m) = \frac{(b-a)^2}{24} f''(\eta)}$$

for some $\eta \in (a, b)$. These formulas refine Hermite–Hadamard: convexity determines the signs, while the curvature f'' determines the magnitudes.

For $f(x) = x^2$ on $[-1, 1]$, the true average is $1/3$, the midpoint value is 0, and the endpoint average is 1. Since $f'' = 2$, the displayed formulas predict gaps $1/3$ and $2/3$, exactly as observed.

§1.3 A sharp reciprocal inequality from one pointwise bound

Assume $0 < m \le f(x) \le M$ almost everywhere on $[a, b]$, and let

$$\mathbb{E}[g] = \frac{1}{b-a}\int_a^b g(x)\,dx.$$

The product $(M - f)(f - m)$ is nonnegative, so

$$f + \frac{mM}{f} \le m + M.$$

Averaging and applying AM–GM to the two nonnegative terms gives

$$2\sqrt{mM\,\mathbb{E}[f]\mathbb{E}[1/f]} \le \mathbb{E}[f] + mM\,\mathbb{E}[1/f] \le m + M.$$

Therefore

$$\boxed{\mathbb{E}[f]\,\mathbb{E}[1/f] \le \frac{(m+M)^2}{4mM}.}$$

This is the integral version of the Kantorovich inequality. Cauchy–Schwarz supplies the reverse inequality $\mathbb{E}[f]\mathbb{E}[1/f] \ge 1$.

The upper bound is sharp: choose $f = m$ on half the interval and $f = M$ on the other half. Then

$$\mathbb{E}[f] = \frac{m+M}{2}, \qquad \mathbb{E}[1/f] = \frac{1/m + 1/M}{2}.$$

The product equals the claimed constant. When continuity is imposed and $m < M$, the sharp constant is approached by continuous approximations of this two-level function.

§1.4 An involution can evaluate a logarithmic integral

Consider

$$I = \int_0^1 \frac{\log(1+x)}{1+x^2}\,dx.$$

Set $x = \tan t$, so $0 \le t \le \pi/4$, and obtain

$$I = \int_0^{\pi/4} \log(1 + \tan t)\,dt.$$

The reflection $t \mapsto \pi/4 - t$ preserves the integration interval. Moreover,

$$1 + \tan\left(\frac{\pi}{4} - t\right) = \frac{2}{1 + \tan t}.$$

Consequently

$$I = \int_0^{\pi/4} \big(\log 2 - \log(1 + \tan t)\big)\, dt = \frac{\pi}{4}\log 2 - I.$$

The exact value follows without expanding any logarithm:

$$\boxed{I = \frac{\pi}{8}\log 2.}$$

The essential move was to find a measure-preserving reflection under which the integrand transforms into a constant minus itself.

The same pairing philosophy proves positivity of an oscillatory example:

$$\begin{aligned}\int_0^{\sqrt{2\pi}} \sin(x^2)\, dx &= \frac{1}{2}\int_0^{2\pi} \frac{\sin t}{\sqrt{t}}\, dt \\ &= \frac{1}{2}\int_0^{\pi} \sin t \left(\frac{1}{\sqrt{t}} - \frac{1}{\sqrt{t+\pi}}\right) dt > 0.\end{aligned}$$

The two half-periods have opposite signs, but the positive one carries the larger weight. This is an elementary instance of oscillatory cancellation with monotone amplitude.

§1.5 Opposite-Point Averages

The reflection $t \mapsto -t$ can turn a symmetric pair of integrals into a derivative test. For f continuous near 0 and $x > 0$, write

$$H(x) = \int_0^x f(t)\, dt + \int_0^{-x} f(t)\, dt = \int_0^x \big(f(t) - f(-t)\big)\, dt.$$

The integral mean-value theorem supplies some $\vartheta_x \in (0,1)$ with

$$H(x) = x\big(f(\vartheta_x x) - f(-\vartheta_x x)\big).$$

What can we say about ϑ_x as the interval shrinks? If f is differentiable at 0 and $f'(0) \neq 0$, then

$$f(t) - f(-t) = 2tf'(0) + o(t), \qquad H(x) = x^2 f'(0) + o(x^2).$$

The remainder estimate is uniform for $0 \le t \le x$, so the mean-value identity gives

$$x^2 f'(0) + o(x^2) = 2\vartheta_x x^2 f'(0) + o(x^2); \qquad \boxed{\vartheta_x \longrightarrow \tfrac{1}{2}}.$$

The nonzero-derivative hypothesis matters. For $f(t) = t^3$, the identity is exactly

$$\frac{x^4}{2} = 2\vartheta_x^3 x^4, \qquad \vartheta_x = 4^{-1/3},$$

a fixed value different from $1/2$. The representative point of an integral is governed by the first nonvanishing odd-order term, not automatically by the midpoint.

§1.6 Two antiderivatives hiding in an arcsine integral

Consider

$$A = \int_0^1 \frac{\arcsin\sqrt{x}}{\sqrt{x(1-x)}}\,dx.$$

The weight looks singular at the endpoints, but it is integrable. Put $u = \arcsin\sqrt{x}$, so $x = \sin^2 u$ and $dx = 2\sin u\cos u\,du$. Thus the apparent singularity disappears:

$$A = 2\int_0^{\pi/2} u\,du = \boxed{\frac{\pi^2}{4}}.$$

There is a second proof worth keeping, because it anticipates beta integrals. The involution $x \mapsto 1-x$ preserves $dx/\sqrt{x(1-x)}$, while

$$\arcsin\sqrt{x} + \arcsin\sqrt{1-x} = \frac{\pi}{2}.$$

Averaging the transformed and untransformed integrals yields

$$2A = \frac{\pi}{2}\int_0^1 \frac{dx}{\sqrt{x(1-x)}} = \frac{\pi}{2}B\left(\tfrac{1}{2},\tfrac{1}{2}\right) = \frac{\pi^2}{2}.$$

This is also the arcsine probability density in disguise: $\left(\pi\sqrt{x(1-x)}\right)^{-1}$ is a probability density on $(0,1)$, and symmetry makes its mean transformed angle $\mathbb{E}[\arcsin\sqrt{X}] = \pi/4$. The same integral has an elementary substitution, a symmetry proof, and a probabilistic interpretation.

§1.7 Why a cosine integral leads to the Poisson kernel

For $a > |b|$ with $a > 0$, the family

$$J(a,b) = \int_0^\pi \frac{dx}{a + b\cos x}$$

has a surprisingly clean value. The substitution $t = \tan(x/2)$ gives

$$\cos x = \frac{1-t^2}{1+t^2}, \qquad dx = \frac{2\,dt}{1+t^2}, \qquad 0 < t < \infty.$$

Consequently

$$\begin{aligned} J(a,b) &= 2\int_0^\infty \frac{dt}{(a+b) + (a-b)t^2} \\ &= \boxed{\frac{\pi}{\sqrt{a^2-b^2}}}. \end{aligned}$$

Changing the interval changes the answer: the integral over $[-\pi/2, \pi/2]$ is instead

$$\int_{-\pi/2}^{\pi/2} \frac{dx}{a+b\cos x} = \frac{4}{\sqrt{a^2-b^2}}\arctan\sqrt{\frac{a-b}{a+b}}.$$

One can check both formulas immediately at $b = 0$.

A more structural derivation explains the radical. Set $\Delta = \sqrt{a^2-b^2}$, $C = (a+\Delta)/2$, and $r = -b/(a+\Delta)$. Then $|r| < 1$ and

$$a + b\cos x = C(1 - 2r\cos x + r^2), \qquad C(1-r^2) = \Delta.$$

Therefore

$$\frac{1}{a+b\cos x} = \frac{1}{\Delta}\,\underbrace{\frac{1-r^2}{1-2r\cos x+r^2}}_{P_r(x)}.$$

The factor P_r is the Poisson kernel for the disk. Its Fourier expansion

$$P_r(x) = 1 + 2\sum_{n=1}^{\infty} r^n \cos(nx)$$

converges absolutely for $|r| < 1$; integrating from 0 to π kills every nonconstant cosine. We recover $J(a,b) = \pi/\Delta$ and see why a rational trigonometric integral is naturally related to harmonic functions on the disk.

§1.8 Variation Bounds from Derivatives

Convexity is not the only source of integral inequalities. Let $f \in C^2([a,b])$ satisfy $f(a) = f(b) = 0$ and $f > 0$ on (a,b). Write $M = f(x_0) = \max f$ for an interior maximizer. The mean-value theorem gives points $\xi_1 \in (a, x_0)$ and $\xi_2 \in (x_0, b)$ with

$$f'(\xi_1) = \frac{M}{x_0 - a}, \qquad f'(\xi_2) = -\frac{M}{b - x_0}.$$

Since $f \le M$ where it is positive, we have (allowing the left side to be infinite)

$$\begin{aligned}\int_a^b \left|\frac{f''(x)}{f(x)}\right| dx &\ge \frac{1}{M}\int_a^b |f''(x)|\,dx \\ &\ge \frac{f'(\xi_1) - f'(\xi_2)}{M} \\ &= \frac{1}{x_0 - a} + \frac{1}{b - x_0} \ge \boxed{\frac{4}{b-a}}.\end{aligned}$$

The middle inequality is the total-variation estimate $\int |f''| \ge |f'(\xi_2) - f'(\xi_1)|$. This is a different kind of argument from Hermite–Hadamard: a function forced to rise from zero and return to zero must pay a definite amount of curvature. For the smooth profile $f(x) = \sin(\pi(x-a)/(b-a))$, the integral is $\pi^2/(b-a)$, which confirms the scale $1/(b-a)$ and shows why the dimensions of the estimate are natural.

§1.9 Wallis integrals become beta integrals

Define

$$I_n = \int_0^{\pi/2} \sin^n x\,dx, \qquad n \ge 0.$$

Integration by parts shows

$$I_n = \frac{n-1}{n} I_{n-2} \quad (n \ge 2), \qquad I_0 = \frac{\pi}{2}, \quad I_1 = 1.$$

Hence the even and odd subsequences are controlled by alternating products of consecutive integers.

There is a useful closed form that does not require treating the two parities separately. With $t = \sin^2 x$,

$$\begin{aligned} I_n &= \frac{1}{2}\int_0^1 t^{(n-1)/2}(1-t)^{-1/2}\,dt \\ &= \frac{1}{2}B\left(\frac{n+1}{2},\frac{1}{2}\right) = \frac{\sqrt{\pi}\,\Gamma((n+1)/2)}{2\Gamma((n+2)/2)}. \end{aligned}$$

The beta function is Euler's integral $B(u,v) = \int_0^1 t^{u-1}(1-t)^{v-1}dt$, and the last equality uses $B(u,v) = \Gamma(u)\Gamma(v)/\Gamma(u+v)$.

The asymptotic behavior has a geometric explanation. Writing $x = \pi/2 - t$, we have

$$I_n = \int_0^{\pi/2} \cos^n t\,dt.$$

For large n, the mass concentrates near $t = 0$, where $\log\cos t = -t^2/2 + O(t^4)$. The resulting Gaussian scaling, justified by Laplace's method, gives

$$\boxed{I_n \sim \sqrt{\frac{\pi}{2n}}.}$$

The recurrence produces Wallis' product for π, while the beta–gamma formula and Gaussian approximation describe the same sequence from special-function and asymptotic viewpoints.

There is one more payoff. Regard $I_s = \int_0^{\pi/2} \sin^s x\,dx$ as a function of a real parameter $s > -1$. Differentiation under the integral sign gives

$$I_s' = \int_0^{\pi/2} \sin^s x \log(\sin x)\,dx.$$

At $s = 1$ this is the logarithmic sine integral. It has an elementary evaluation that checks the parameter method without invoking derivatives of the gamma function:

$$\begin{aligned} I_1' &= \int_0^{\pi/2} \sin x \log(\sin x)\,dx \\ &= \frac{1}{2}\int_0^1 \log(1-u^2)\,du \qquad (u = \cos x) \\ &= \frac{1}{2}\left(\int_0^1 \log(1-u)\,du + \int_0^1 \log(1+u)\,du\right) \\ &= \boxed{\log 2 - 1}. \end{aligned}$$

The parameter s has converted an integral with a logarithm into the derivative of an ordinary beta integral. This is a useful general strategy for moments and normalizing constants in probability.

Averaging, reflection and recurrence all reduce an integral to simpler data: a convex bound, an invariant pair of points, or a relation between neighboring exponents. Here the special-function language grows directly from calculations one can still check by elementary substitutions.